

ML Foundations — Day 3: Feature Engineering, CV & Model Comparison

Dataset: Adult / Census Income (UCI, via OpenML) | Qasim | random_state = 42

Engineered Feature List

Feature	Type	Rule	Signal
age_bucket	numeric_bucket	18-25 / 26-35 / 36-50 / 51+	MI (one-hot) = 0.0273
hours_bucket	numeric_bucket	<30 / 30-40 / 41-50 / 50+	MI (one-hot) = 0.0261
capital_gain_flag	binary_flag	1 if capital-gain > 0	$P(>50K 0)=0.194$, $P(>50K 1)=0.696$
capital_gain_log	log_transform	$\log(1 + \text{capital-gain})$	MI = 0.0645
higher_education	binary_flag	1 if education-num ≥ 13	$P(>50K 0)=0.156$, $P(>50K 1)=0.470$
education_x_hours	interaction	education-num $\times$ hours-per-week	MI = 0.0728
age_x_hours	interaction	age $\times$ hours-per-week	MI = 0.0663
marital_married	binary_flag	1 if marital-status = Married-civ-spouse	$P(>50K 0)=0.049$, $P(>50K 1)=0.446$

All 8 features are computed row-wise from raw columns only (no target information, no cross-row aggregation), so they carry no leakage risk. Binary flags with the largest $P(>50K|1)$ vs $P(>50K|0)$ gap — marital_married (0.446 vs 0.049) and capital_gain_flag (0.696 vs 0.194) — show the strongest univariate signal.

Cross-Validation Comparison (5-fold, training split only)

Model	ROC AUC (mean $\pm$ std)	F1 (mean $\pm$ std)	Accuracy	Precision	Recall	CV time
Logistic Regression	0.9114 $\pm$ 0.0032	0.6679 $\pm$ 0.0047	0.8557	0.7436	0.6064	2.3s
Random Forest	0.9006 $\pm$ 0.0027	0.6628 $\pm$ 0.0051	0.8506	0.7210	0.6134	15.3s
HistGradientBoosting	0.9262 $\pm$ 0.0036	0.7031 $\pm$ 0.0082	0.8698	0.7739	0.6444	9.5s

HistGradientBoosting leads on every metric (ROC AUC 0.926, F1 0.703) and is also fast (9.5s CV). Boxplots of the 5 fold scores show HistGradientBoosting's ROC AUC fold scores tightly clustered above both other models with no overlap in the interquartile range, while Logistic Regression and Random Forest boxes sit close together and do overlap — visually consistent with the statistical test below.

Statistical Test Results

Top 2 models by mean ROC AUC: **HistGradientBoosting** (folds: 0.9259, 0.9329, 0.9256, 0.9246, 0.9219) vs. **Logistic Regression** (folds: 0.9141, 0.9157, 0.9090, 0.9112, 0.9068). Paired t-test: $t = 14.80$, **$p = 0.0001$** . Wilcoxon signed-rank (non-parametric cross-check): $p = 0.0625$ (the smallest possible p-value achievable with only 5 paired samples, since Wilcoxon has very limited resolution at this sample size). **Interpretation:** the t-test result is a strong signal that HistGradientBoosting's advantage is not noise — every single fold favored it, by a consistent margin (~ 0.011 – 0.017 ROC AUC each). The Wilcoxon p just above the conventional 0.05 threshold reflects the test's low power with $n=5$, not real uncertainty about the direction of the effect; given the t-test and the fold-by-fold consistency, this is treated as a practically meaningful, not chance, difference.

Feature Importance Insights

Random Forest top importances: fnlwgt (0.124), age_x_hours (0.088), age (0.078), education_x_hours (0.061), capital-gain (0.051), marital_married (0.046), marital-status_Married-civ-spouse (0.045), education-num (0.044), capital_gain_log (0.039). The two engineered interaction terms rank 2nd and 4th overall, ahead of every raw one-hot category — handing the tree these interactions directly clearly added value beyond what it could infer alone.

Logistic Regression top positive coefficients include capital-gain (3.43), marital-status_Married-AF-spouse (1.51), native-country_Ireland (1.13), **capital_gain_log (0.99, 4th overall)**, education-num (0.90), and **marital_married (0.86, tied 6th with marital-status_Married-civ-spouse)** — both engineered features land in the global top 10. **Notable finding:** capital_gain_flag has a strongly negative coefficient (-4.70), the single largest-magnitude coefficient in the model, despite raw capital gain being strongly positive. This is a multicollinearity artifact: capital_gain_log is already 0 exactly when capital-gain is 0, so the flag and the log feature are highly redundant, and the flag's coefficient partly corrects the log feature's behavior at zero rather than reflecting a genuine negative income effect.

Overall, engineered interaction and log features clearly mattered — for Random Forest, they add signal beyond what raw features alone provide even though trees can in principle learn interactions; for Logistic Regression, which cannot model interactions without them being engineered explicitly, `capital_gain_log` and `marital_married` both rank in the global top 10.

Feature Selection Decision

Full engineered feature set (119 columns post one-hot): ROC AUC 0.9114 ± 0.0032 , 2.3s CV time. SelectKBest (mutual information, $k=59$, roughly half): ROC AUC 0.9091 ± 0.0034 , 51.2s CV time (the MI scoring pass dominates the cost, not the downstream fit). The ~ 0.0023 ROC AUC drop is well within one fold's standard deviation — selection neither helped nor hurt meaningfully, and cost far more compute. **Decision: keep the full engineered feature set for Day 4's hyperparameter tuning**, with one targeted exception — drop `capital_gain_flag` specifically (keep `capital_gain_log`), since Task 4 showed the two are highly collinear and the flag adds interpretive noise to Logistic Regression without adding predictive signal Random Forest couldn't already extract from `capital_gain_log` alone. Row-wise-only feature engineering (no target-encoding, no cross-row aggregation) is kept unchanged into Day 4, since it already captured the useful signal without leakage risk.